

Specification, version 3

What we are building, what we have agreed, what is still open, and how the numbers are worked out

24 September 2026 | Prepared by Dupleix Institute for the Malawi Agricultural and Industrial Investment Corporation plc (MAIIC)

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Status: working document for the meeting of 25 September 2026. Replaces the consolidated specification of 20 August 2026 as the single point of reference. It does not delete anything from that document; where this one is silent, the August document still applies.

0. How to read this document

This specification is written for three readers at once: the CFO, who has to be confident that the numbers will stand up to Deloitte; the ICT manager, who has to supply the data and host the system; and the developers, who have to build it. So every section starts with the plain-language version and only then goes into detail. Words that a non-specialist might not know are explained in the glossary at the end (section 15).

Three conventions are used throughout:

- **"Agreed"** means a decision that has been taken, by whom and when. Section 3 lists them. They are not re-opened here.
- **"Not yet agreed"** means a choice that still has to be made. Section 4 lists them, each with the options and our recommendation. Nothing in section 4 is assumed anywhere else in the document without saying so.
- **"Proven"** means a fact that was tested against MAIIC's own data and reproduced. Where a figure is proven, the file and the count are given so that anyone can repeat the test.

One phrase is used deliberately every time: **the spread added to the prime rate (margin)**. In both of MAIIC's system manuals the bare word "margin" means something else (a collateral haircut in the loan origination manual, page 63; a grace concept in the core banking manual, pages 23 and 31). To avoid any confusion we never use the bare word.

1. Background, explained from the beginning

1.1 What an effective interest rate is, and why IFRS 9 insists on it

When MAIIC lends money, the borrower signs an offer letter that states an interest rate, a repayment pattern and, usually, an arrangement fee and legal fees that are deducted before the money is paid out. The rate on the letter is the **contractual rate**. But the borrower did not receive the full loan: fees came off the top. So the true return MAIIC earns on the cash it actually parted with is a little higher than the contractual rate. That true return is the **effective interest rate**, or EIR.

IFRS 9 (Appendix A) defines the EIR as the rate that exactly discounts the expected future cash payments over the expected life of the loan to the amount MAIIC actually advanced, net of fees that are an integral part of the loan. Interest income is then recognised at that rate on the loan's **amortised cost** (the amount advanced, plus interest recognised, minus cash received), not at the contractual rate on the outstanding principal. The difference between the two is small on any single month but it changes the pattern of income across the life of the loan, and it is what the auditors test.

Three IFRS 9 paragraphs do most of the work in this engine:

Paragraph	What it says, plainly	Where it bites for MAIIC
Appendix A (definition)	Use contractual cash flows, ignore expected credit losses, include integral fees	Every loan
B5.4.5	For a floating-rate loan, when the reference rate changes, re-estimate the future cash flows and carry on at the new rate. No catch-up adjustment	The 39 or so MAIIC loans that follow the Reserve Bank prime rate, 26 times since 2020

Paragraph	What it says, plainly	Where it bites for MAIIC
B5.4.6	If the estimate of cash flows changes for any other reason (for example, arrears or a changed repayment plan), recompute the amortised cost at the original EIR and take the difference to income now	Every loan in arrears; every restructured loan
5.4.3	When a loan is modified but not derecognised, keep the original EIR and book a modification gain or loss	The restructured loans
B3.3.6 (by analogy)	The "10 percent test": if the present value of the new cash flows differs from the old by 10 percent or more, treat it as a new loan	Deciding whether a restructure is a modification or a new loan
5.4.1(b)	Once a loan is credit-impaired (Stage 3), calculate interest on the net carrying amount (after the loss allowance)	Stage 3 loans
5.5.11 and B5.5.37	30 days past due is the backstop for a significant increase in credit risk; 90 days is the backstop for default	Stage allocation, which the existing ECL module already does

1.2 What MAIIC has today

MAIIC's core banking system is **E-Banker**, from Virtual Galaxy, live since about October 2024 (the exact date is one of the questions for the meeting). Loans are originated in a separate **Loan Origination System (LOS)** and then pushed into E-Banker. E-Banker calculates and posts contractual interest every month; it does not calculate an EIR, an amortised cost or an expected credit loss. Dr Thom's own December 2025 assessment states this, and our search of the vendor's documentation found no IFRS 9 or EIR capability.

The IFRS 9 expected credit loss (ECL) calculation runs in the **MAICC-IFRS9** system that Duplex built for MAIIC. That system already holds the monthly loan book, the client records, the staging rules and the ECL engine. The EIR engine described here is a module inside that same system (agreed decision D1), so that the ECL module and the EIR module read the same loan book and the same stage for every loan.

1.3 The three doors

The EIR enters IFRS 9 three times, and the engine has to serve all three:

1. **Measurement (door 1):** the amortised cost of each loan at each month-end.
2. **Revenue (door 2):** the interest income recognised each month, and the difference between that and what E-Banker posted contractually.
3. **Impairment (door 3):** the rate at which expected cash shortfalls are discounted in the ECL calculation. The ECL module used to use a placeholder; since August 2026 it reads the solved EIR.

1.4 What has happened so far (the short history)

Date	Event
21 Jul 2026	Award letter from MAIIC's Internal Procurement Committee
27 Jul to 18 Aug 2026	Schema, import layer, solver, impairment rewiring and revenue engine built on branch <code>eir_revenue_recognition</code>
31 Jul 2026	Contract Effective Date (DUP/MAIIC/IFRS9/2026, USD 30,000, four milestones)
3 Aug 2026	Extracts A, B and C received from MAIIC

Date	Event
4 Aug 2026	Specification v1
19 Aug 2026	Contract signed; trial-balance corpus and GL spools received
20 Aug 2026	Consolidated specification (v2.3)
19 Aug to 2 Sep 2026	The September build: trial-balance corpus, original-schedule governance (draft to approved version 1), date-sensitive solver with controlled reopening, time-phased ECL discounted at the EIR, standing fee rulebook, GL accrual base derived from the data
3 and 10 Sep 2026	Source-screen walkthroughs with MAIIC and the vendor: where each EIR input lives (LOS or E-Banker), the two schedules, fees posted by Finance, EMI-only repayment, restructure as a new sub-account
10 Sep 2026	Consolidated specification v2.5: Path E (source screens), the floating-rate reset design (Phase 5.1), the data request register (Appendix E)
11 Sep 2026	Consolidated Information Request to MAIIC: 15 numbered items
17 Sep 2026	Interest Rate Change History file received (dates later found to be corrupted in Excel, not in E-Banker)
21 Sep 2026	E-Banker and LOS user manuals received (request 14 closed)
22 to 24 Sep 2026	Teaching workbook, extracts research, field reference, reconstruction test, Extract B check, loan-book date scan
25 Sep 2026	Progress meeting with Dr Thom and Barry Makumba

2. Where this document comes from: the references

Everything in this specification traces to one of the sources below. They are cited by their short name in square brackets.

Short name	Document	Where
[Spec v1]	MAIIC EIR and Revenue Recognition Engine, Technical Specification, 4 Aug 2026	OneDrive 3. Project Execution\specs\MAIIC_EIR_Revenue_Recognition_Engine_Spec_v1_2026-08-04.md
[Spec v2]	Consolidated Technical Specification, v2.5 of 10 Sep 2026 (v2.3 of 20 Aug in OneDrive)	docs/MAIIC_EIR_Revenue_Recognition_Engine_Spec.md in the repository; the OneDrive copy is v2.3
[Inventory]	Source Screen Field Inventory from the 3 and 10 Sep 2026 walkthroughs	2. Documents from clients\MAIIC EIR - Source Screen Field Inventory.docx
[Analysis]	Branch inspection report, 24 Sep 2026	docs/EIR_Branch_Analysis_2026-09-24.md in the repository; copy in the meeting pack

Short name	Document	Where
[Explained]	The engine explained, plain-language companion, 5 Aug 2026	same folder, ..._Explained.md/.pdf
[Build]	EIR Build (phases and acceptance bar) and Development of EIR (build log)	docs/EIR_Build.md , docs/Development_of_EIR.md
[Research]	MAIIC Data Extracts, Research Report, 24 Sep 2026	specs\MAIIC_Extracts_Research_Report_2026-09-24.md
[Fields]	E-Banker and LOS field reference with manual screenshots, 24 Sep 2026	specs\MAIIC_EIR_Field_Reference_2026-09-24.pdf
[Workbook]	Field reference workbook: every extract column, its manual name and page, which request asked for it	meeting pack, MAIIC_Extracts_Field_Reference_Workbook_2026-09-24.xlsx
[Recon]	Reconstruction test: the ten offer letters against the GL, 24 Sep 2026	Amortisation Analysis\offer_letter_reconstruction_test_2026-09-24.csv
[ExtB]	Extract B against four offer letters, 24 Sep 2026	Amortisation Analysis\extract_b_offer_letter_check_2026-09-24.md/.csv
[DateScan]	Loan-book date scan, 24 Sep 2026	specs\loan_book_date_scan_2026-09-24.md
[PLR]	Repaired reference-rate series and the 20 rows for Barry to confirm	Reference Rates\plr_reference_rates_corrected.csv , plr_dates_for_barry_to_confirm.csv
[Teach]	IFRS 9 floating-rate EIR teaching workbook (10-year loan, fees, three missed instalments, year-6 restructure, 28 integrity checks)	Documents\IFRS9_Floating_Rate_EIR_Illustration.xlsx
[Manual]	E-Banker user manual (pages cited as "E-Banker p.NN")	MAIIC folder, Ebanker MANUAL.pdf
[LOS]	Loan Origination System manual (pages cited as "LOS p.NN")	MAIIC folder, LOS MANUAL.pdf
[Request]	Consolidated Information Request of 11 Sep 2026 (items 1 to 15)	MAIIC folder, Email requests ... UPDATED (to MAIIC).html

Short name	Document	Where
[Contract]	Agreement DUP/MAIIC/IFRS9/2026, signed 19 Aug 2026	1. Engagement Contracting\Final Contract
[Deck]	Progress deck for the 25 Sep 2026 meeting (20 slides with presenter notes)	meeting pack, MAIIC_EIR_Progress_and_Data_Requests_2026-09-25.pptx

3. What we have agreed

Each decision below has been taken. The date and the person are given so that nobody has to re-derive it. Where a decision was taken by Edward Mazibuko on 24 September 2026 in answer to the fourteen design questions, it is marked "EM 24 Sep".

3.1 Scope and shape of the build

#	Decision	By	Why
D1	The EIR engine is a module inside the existing MAICC-IFRS9 system , on branch <code>eir_revenue_recognition</code> . There is no separate repository and no separate application.	EM 24 Sep	One loan book, one stage per loan, one login for the finance team; the ECL module already reads the loan book
D2	Users are the MAIIC finance team . The system is deployed into MAIIC's own environment . Runs are month-end only .	EM 24 Sep	Matches the contract deliverables (configured client system, source code, training)
D3	Arrears and restructuring are in version 1 . They are not deferred to a later phase.	EM 24 Sep	Both are common in the book and both change the EIR calculation
D4	The third moratorium shape (pay capital, defer interest) is not built . E-Banker's Moratorium Type has exactly two options, "Principle Only" and "Both (Interest + Principle)" (LOS p.76), and no loan book in 33 months shows a third shape.	EM 24 Sep, confirmed by [Manual]	Do not build what the source system cannot express
D5	FinES concessional loans (fixed 10 percent, funded by the Reserve Bank) are calculated on their stated terms . No day-one fair-value adjustment.	EM 24 Sep	Keeps the engine to what the auditors have asked for
D6	The engine does not post to the general ledger . It proposes journals; Finance posts them in E-Banker.	[Spec v2] s.14	Scope boundary agreed in August
D7	MAIIC supplies the whole population of loans, with the spread added to the prime rate (margin), by CSV import or by integration. The ten offer letters are a test sample, not the data source.	EM 24 Sep	"I only asked for 10 samples. We expect the client to give us the 100 percent population"

3.2 How the calculation works

#	Decision	By	Why
D8	One EIR per rate period. A new child table <code>eir_rate_periods</code> holds the EIR that applied from each reset date. <code>contract_eir</code> stays the origination record and is never overwritten.	EM 24 Sep	B5.4.5: a floating loan gets a fresh EIR at every reference-rate change; the history must survive. [Spec v2] s.7.5 chose the spread-lock method (new EIR = new contractual rate + the fee spread locked at origination) with a full re-solve as the test oracle; each rate period stores both and flags any disagreement beyond solver tolerance
D9	Day count is actual/365, compounding monthly , as the native convention. 30/360 is available as a governed alternative.	EM 24 Sep; proven in [Recon]	E-Banker's own postings reproduce to the cent on this basis (section 7.7)
D10	A capitalising moratorium compounds monthly : the balance rises each month by exactly the interest posted.	proven in [Recon]	Observed on every moratorium loan in the sample
D11	Stage 3 interest is recognised on the net carrying amount (IFRS 9 5.4.1(b)) by default. Governable.	EM 24 Sep	Inherits the standard's rule; the ECL module supplies the stage
D12	The reference rate is the Reserve Bank prime lending rate (PLR) . The rate typed in the body of an offer letter is the reference rate; the rate on the repayment schedule is the reference rate plus the spread added to the prime rate (margin); the schedule rate is the contractual rate. FinES loans are always fixed; MAIIC's own loans are usually PLR-linked.	EM 24 Sep, confirmed on 5 of 6 letters in [Recon]	The two rates on the letters disagreed in 6 of 10 and this explains why
D13	The spread added to the prime rate (margin) is derived by the engine as the loan-book rate minus the PLR in force at the month-end, and checked for constancy across months. A margin supplied by MAIIC is used as a cross-check, not as a dependency.	EM 24 Sep, on the 8-of-8 evidence in [Research]	Removes the engine's dependence on a field E-Banker does not hold
D14	Actual cash received comes from the monthly Loan Book Report by default: this month's cash is the increase in the cumulative <code>Repayments</code> column, which ties exactly to the fall in <code>Outstanding Balance</code> . Extract B (the transaction ledger) is an optional second source.	EM 24 Sep	The loan book is already supplied monthly; Extract B is not
D15	Where instalments are not level (arrears, moratoria, restructures), the EIR is solved by internal rate of return on the actual expected cash flows , never by a closed-form annuity formula.	[Teach], 55 basis-point error demonstrated	A level-annuity formula understates the EIR whenever the stream is not level

#	Decision	By	Why
D16	E-Banker's own codes are stored verbatim (Interest Policy, Floating Flag, Loan Interest Cal. Base On, Installment Based On, Moratorium Type, status codes) and the engine's internal categories are derived from them. Nothing is inferred from the Floating Flag alone.	[Research] s.6	RATE_BASIS in Extract A turned out to be the Floating Flag, not the PLR link

3.3 Governance, data and control

#	Decision	By	Why
D17	A Governance Centre holds every calculation convention as a setting. Each setting has options, a default equal to Duplex's recommendation, an effective date, an approver, and an audit log of who changed what, from what, to what, and why. A change never silently restates a locked period.	EM 24 Sep	MAIIC staff operate the system themselves; the auditor has to see every convention and every change
D18	Import is four files : A, the contract master (with the spread added to the prime rate (margin)); B, the transaction ledger (optional); C, the reference-rate series; D, repayment schedules as issued (optional). Plus the monthly Loan Book Report, which the ECL module already loads. Single and bulk import, with a column-mapping screen.	EM 24 Sep	Matches what MAIIC can produce today
D19	The reference-rate series is loaded from the repaired file [PLR] with ISO dates (yyyy-mm-dd) . The importer rejects any date column that is mixed-type or ambiguous; it never guesses day against month.	[Research], 1,134 wrong dates found	The Excel locale problem must not reach the engine
D20	The acceptance test for interest is the reconstruction in [Recon]: on a clean month the engine's contractual interest must equal E-Banker's posting to the cent; every difference must be explained by a named cause (late disbursement, catch-up posting, mid-month tranche).	[Recon]	This is what "reconcile within the agreed tolerance" in the contract will mean in practice
D21	Every build on the branch starts and ends with the EIR test suite green on the in-memory test database, never against the live development database. Baseline confirmed on 24 Sep 2026: 124 tests, 512 assertions, all passing.	[Spec v2] s.12; repo check 24 Sep	A RefreshDatabase test once wiped the dev database; the baseline is now proven, so a regression will show

3.4 Facts established with MAIIC and the vendor at the September walkthroughs

These were confirmed on 3 and 10 September 2026 and recorded in [Spec v2] s.3.6 and [Inventory]. They are treated as settled.

#	Fact	Consequence for the engine
F1	LOS holds the rate build-up, the fees and the moratorium configuration; on acceptance it creates the account in E-Banker, which owns disbursement, transactions, accrual and the amortisation table. The LOS application number and process reference are the only join, and they reach us in no extract.	Section 5.1 asks for both identifiers

#	Fact	Consequence for the engine
F2	There are two schedules. The LOS origination schedule is the one the customer signs; it cannot be downloaded after commit, is not visible to Credit, and prints only as PDF. The E-Banker EMI chart exports to CSV but omits the opening (disbursement) row; MAIIC has a change request open with the vendor to add it.	File D is optional (5.4); the engine generates version 1 from terms and reconciles to E-Banker (7.7)
F3	The EMI chart marks a missed instalment as paid and keeps amortising.	The chart is evidence of resets, never an actuals feed; cash comes from the loan book (D14)
F4	Migrated loans (written before E-Banker) were migrated at carrying amount only and have no origination schedule in either system; their pre-migration Excel schedules sit with Finance.	The take-on schedules of 31 Oct 2024 are the only full-life vector for that cohort
F5	FinES is fixed at 10 percent for life; MAIIC Industrial and Agricultural float with the Reserve Bank rate, repriced whenever it changes (roughly every three to four months). E-Banker holds a table of interest rates by effective date that can be spooled.	File C (5.3) is that table; the repaired copy is [PLR]
F6	Arrangement and legal fees are captured manually in E-Banker by Finance at disbursement, read off the offer letter, deducted from the amount advanced, and posted to GL 4873 (arrangement), 4871 (legal) and 4872 (consultancy). No screen or schedule carries the fee as a cash flow.	Fees need their own feed (O14); the initial net investment is drawn amount less fees
F7	Only EMI repayment is in use; straight-line and bullet are configurable but not used.	The level-instalment generator covers the whole current book
F8	A restructure increments the sub-account number on the same main account (1, 2, 3) and E-Banker can produce a per-account audit report. Extract A nonetheless shows sub-account 1 and a blank restructure date on every row.	Lineage exists at source and is lost on export; the ask is the Reschedule Report and the sub-account sequence

4. What has not yet been agreed

Each item below is a real choice. The options are listed with Dupleix's recommendation first. None of them blocks the build of the engine's core, but each one changes a number somewhere, so each must be settled and recorded in the Governance Centre before the first locked run.

#	Open choice	Options (recommendation first)	Who decides	What it depends on
O1	Margin derivation basis. Where does the spread added to the prime rate (margin) come from when the loan-book rate and the LOS rate disagree?	(a) LIVE_AT_DRAWDOWN : loan-book rate minus the PLR in force on the disbursement date, then held constant; (b) AS_CAPTURED : LOS reference rate plus the margin as typed at origination; (c) SUPPLIED : a margin column from MAIIC	Dr Thom, after Barry answers the refresh-button question	Whether the LOS reference-rate field is a stamped snapshot (refreshed by a button) or a live link. Ebenezer Midian was approved on 8 Aug 2025 with 25.10 percent typed, but the PLR had been 25.30 percent since early July

#	Open choice	Options (recommendation first)	Who decides	What it depends on
O2	When a PLR change takes effect in the calculation.	(a) From the Reserve Bank effective date, pro rata by days within the month; (b) from the next month-end; (c) from the next instalment date	Dr Thom	How E-Banker itself applies a mid-month change. The month-end rate moves with the PLR; whether the posting is split within the month is not yet proven
O3	Rate source precedence when the sources disagree.	(a) E-Banker Interest Policy code, then observed behaviour, then product family; (b) observed behaviour first; (c) product family only	Dupleix, confirmed by Dr Thom	Whether MAIIC sends Interest Policy per account (new ask 1)
O4	Treatment of accounts with Interest Policy = Manual .	(a) Read the rate month by month from the loan book and treat each change as a reset; (b) treat as fixed until told otherwise	Dr Thom	Which accounts are Manual (Audit Trail Report, Menu ID 692)
O5	Repayments counter resets to zero (23 resets in nine months, up to MWK 677 million).	(a) Treat a reset as a discontinuity and take that month's cash from Extract B; (b) treat as a settlement and rebook; (c) ask MAIIC per case	Barry to explain the cause; Dr Thom to decide	What causes a reset: settlement, restructure or re-booking
O6	Interest basis on partly drawn facilities. E-Banker can charge on the sanctioned amount or on the balance (E-Banker p.30), and the instalment likewise (p.22). On Lake Malawi Aquaculture the two bases differ by MWK 22 million a month.	(a) Read the flag per loan from E-Banker and follow it; (b) assume balance-wise everywhere and flag exceptions from the reconciliation	Barry to supply the flags	New ask 1 (the six scheme settings per account)
O7	Re-basing the LOS schedule. [ExtB] shows the offer-letter schedules start accrual on the 1st of the month while E-Banker starts on the disbursement day, and two of four sample loans are charged a different rate in E-Banker from the letter (31 vs 25.1; 33 vs 25.3 percent).	(a) Generate the expected cash flows from the core disbursement date and the core rate, and keep the LOS schedule as a reference only; (b) use the LOS schedule as issued and book the difference as a modification	Dr Thom, with an explanation from Barry of why the core rate differs	Why Ebenezer Midian and JAT Group carry a different rate in the core system
O8	The fourteen-month gap (Nov 2024 to Nov 2025): no core-banking loan book exists for these months.	(a) MAIIC re-runs the Loan Book Report for each month (it has an AsOn Date parameter); (b) use the GL-wise Interest Posted Report monthly from Oct 2024 for the interest side and Finance's Excel books for balances, disclosed as a limitation; (c) start the EIR history at Dec 2025 and disclose	Dr Thom	Whether the report can be run for past dates

#	Open choice	Options (recommendation first)	Who decides	What it depends on
O9	Integration versus CSV, permanently.	(a) Extend the Loan Book Report (Menu ID 3868) with the missing fields so the monthly file is the integration; (b) a scheduled database view; (c) an API	Dr Thom (authorises the vendor change request)	Vendor estimate for the change
O10	Which GL account absorbs the EIR-versus-contractual true-up (needed before the engine can propose journals).	(a) A dedicated "EIR adjustment" income account; (b) the existing interest-income account per product	Dr Thom	[Spec v2] open item 16
O11	Reconciliation tolerance.	(a) 100 basis points on the annual EIR and MWK 1 per account-month on interest; (b) an amount threshold agreed with Deloitte	Dr Thom and Deloitte	[Spec v2] open item 11
O12	Deloitte export format.	(a) The Summary-tab shape; (b) the full 23-tab worked example	Kundai, Dr Thom, Deloitte	[Spec v2] open item 17
O13	Phase 0 sign-offs still pending: conventions memo; keyman insurance as an integral fee or not; Nascomex preference-share classification; staging rebuttal.	Sign as drafted	Dr Thom	Unchanged since August
O14	Fee attribution per loan. The GL proves the fees exist (accounts 4871, 4873) but the per-facility allocation is missing; the 2026 NAME column was hand-typed.	(a) Fees as a column set on the extended Loan Book Report; (b) a separate fee extract per posting with the account number; (c) an interim mapping table by Dupleix, disclosed	Barry	Request 4 of 11 Sep, still the largest gap
O15	Roles and permissions for the EIR screens and the auditor downloads.	Reuse the ECL module's role set with three new permissions: view, run, export	Wadzanai, Kundai	[Spec v2] open item 12
O16	Modification threshold.	(a) 10 percent by analogy to B3.3.6; (b) a lower internal threshold	Dr Thom	Whether MAIIC wants a stricter policy than the standard
O17	Reset or modification? MAIIC can vary the spread at its own option where the contract permits. A change linked to the Reserve Bank rate is a B5.4.5 reset; a negotiated rate cut for a struggling borrower can be argued as a 5.4.3 modification with a gain or loss.	(a) Rate moves within the contract are resets; anything negotiated outside the contract is a modification, stated in the accounting policy note; (b) treat every rate change as a reset	Dr Thom, with Deloitte's written confirmation before the first reset is booked	[Spec v2] open item 27
O18	Maker-checker on schedule approval. Version 1 schedules now move from draft to approved, but one person can do both.	(a) Require a second person, as for fee classification and the EIR lock; (b) leave as is for the first run	Dr Thom	[Spec v2] s.12 phase 3.5

5. The data the engine needs, file by file

The engine runs on five inputs. Four are files MAIIC sends (D18); the fifth, the monthly Loan Book Report, the ECL module already loads every month. Every file is loaded through the same column-mapping screen, so a column can be renamed at MAIIC's end without a code change at ours. Every date must be written year first (2026-09-24). A file with an ambiguous date column is rejected with a message naming the column and the first bad row (D19).

5.1 File A: the contract master (one row per loan account, mandatory)

This is Extract A as MAIIC already produces it (36 columns, 181 accounts) plus the fields the manuals show E-Banker holds but the extract does not carry. The plain-language reason for each addition is in [Workbook], sheet "Fields We Are Asking For".

Field	Required	Format	What the engine does with it
LOAN_ACCOUNT_NUMBER + SUB_ACCOUNT_NO	yes	15-digit account, integer sub-account	The key. A restructure creates a new sub-account; the pair identifies one contract version
CUSTOMER_NAME , CUSTOMER_ID	yes	text	Display and the join to the ECL client record
PRODUCT_CODE / loan type, SCHEME_CODE	yes	codes	Scheme carries the six settings below; product family is the fallback prior for rate behaviour
INTEREST_POLICY	yes (new)	one of N M L P S U R W F (E-Banker p.19)	P = reprice at every PLR change; F = never; M = read the rate from the loan book each month (O4)
FLOATING_FLAG	yes (new)	N F I B (p.19-20)	Stored for cross-check only; never used alone to decide repricing (D16)
LOAN_INTEREST_CALC_BASE	yes (new)	N S B (p.30)	S charge interest on the sanctioned amount; B on the balance (O6)
INSTALLMENT_BASED_ON	yes (new)	sanction / disbursement (p.22)	Which amount the instalment is sized on
MORATORIUM_TYPE , MORATORIUM_MONTHS , GRACE_PERIOD_MONTHS	yes	"Principle Only" / "Both"; integers	The shape and length of the moratorium, measured from the first disbursement (p.31)
REPAYMENT_FREQUENCY , TYPE_OF_REPAYMENT , EMI_CALC_TYPE	yes	monthly / quarterly / annual; EMI or equal principal; E P F (p.30)	Shape of the contractual schedule
APPROVED_AMOUNT , DISBURSED_AMOUNT	yes	MWK, 2 dp	Undrawn commitment = approved minus disbursed
LOAN_START_DATE , INTEREST_START_DATE , FIRST_INSTALLMENT_DATE , MATURITY_DATE	yes	ISO dates	Schedule anchors. The core system starts interest on the disbursement day, not the 1st (O7)

Field	Required	Format	What the engine does with it
INTEREST_RATE (contractual, all-in)	yes	percent, 2 dp	The rate E-Banker is charging today
REFERENCE_RATE , SPREAD_OVER_PRIME (margin)	preferred	percent	Cross-check against the derived spread (D13)
ACCOUNT_STATUS	yes	code	Exclude closed accounts; status codes H and F still to be explained
LOS_APPLICATION_NO , LOS_PROCESS_REF	preferred	text	The only link back to the offer letter
RESTRUCTURE_DATE , PREDECESSOR_SUB_ACCOUNT	when restructured	ISO date, integer	Lineage for modification accounting
CURRENCY	yes	MWK	Single currency today; stored for completeness

5.2 File B: the transaction ledger (one row per posting, optional)

Extract B as produced, re-supplied with ISO dates and a debit/credit indicator (the posting screen on E-Banker p.58 has one; the extract does not). Used for three things only: to fill the months where the loan-book Repayments counter resets (O5); to date disbursements within the month; and to separate principal from interest in a repayment, which the extract currently estimates rather than stores. It is optional because D14 takes cash from the loan book.

5.3 File C: the reference-rate series (one row per rate change, mandatory)

Field	Format	Rule
INDEX	text, PLR	One series per index; only PLR exists today
EFFECTIVE_DATE	ISO date	Strictly increasing within an index; the importer refuses a file with any break
RATE	percent, 2 dp	12.00 to 25.40 observed 2020 to 2026
SOURCE_ROW , AS_DELIVERED , INTERPRETATION	text	Audit trail: what the file said and how it was read. The repaired file [PLR] carries 48 rows, 20 of them with day and month swapped back, and 26 genuine rate changes

5.4 File D: repayment schedules as issued (one row per instalment, optional)

The LOS schedule printed on the offer letter: due date, instalment, principal, interest, balance. Loaded as `schedule_version = 1` and kept unchanged for ever. If it is not supplied, the engine generates version 1 from File A (`schedule_source = GENERATED`) and says so. [ExtB] shows the issued schedule is not the cash flow the core account produces (O7), so even when supplied it is the reference, not the expectation. The walkthroughs (F2, F3) settled that the LOS schedule cannot be exported after commit and that the E-Banker EMI chart, which can, omits the opening row and marks missed instalments as paid.

5.5 The monthly Loan Book Report (Menu ID 3868, mandatory, already loaded)

29 columns today: account, customer, National ID, value and maturity dates, tenure with the moratorium tag, moratorium months, interest rate, approved, disbursed, not yet disbursed, principal, collateral, interest

to date, repayments (cumulative), carrying amount, overdue split, four ageing buckets, outstanding balance. The engine reads from it, every month: the rate in force (for the spread and for Manual accounts); cash received (the increase in `Repayments`); arrears (the overdue columns and buckets); the undrawn commitment; and the stage (through the ECL module). [DateScan] confirms this family carries no corrupted dates: 2,018 dates checked, 0 wrong.

If Virtual Galaxy adds the fields in 5.1 to this report (O9, recommended), File A becomes unnecessary and the monthly file is the whole integration.

5.6 Standard E-Banker reports the engine can use as evidence

Not inputs to the calculation, but sources the engine's reconciliation and audit pack can cite: Audit Trail (Menu ID 692, the rate change history from the system of record); Reschedule Report (2839, the restructure population); IFRS9 Detail Report (3863, compounding frequency, instalment dates, the arrears split); GL-wise Interest Posted Report (interest per loan account per month, the source of Extract C); Loan Inquiry (542, the on-demand EMI chart for a single account). The four questions to ask about each are on [Deck] slide 12.

5.7 Validation rules applied on import

1. Dates: ISO only; a column that mixes text and dates, or where any value would parse both ways, is rejected.
2. Amounts: numeric, 2 decimal places, non-negative unless the row is a reversal.
3. Keys: every File B, C and D row must match a File A account, or it is quarantined and listed.
4. Rates: the derived spread over prime must be constant per account across months to within 0.15 percentage points; accounts that drift are quarantined for review, not silently accepted (about 12 of 35 in the first pass).
5. Sums: the principal in File D must equal `DISBURSED_AMOUNT` to within 1 percent.
6. Reference rates: strictly increasing effective dates; no duplicate dates.
7. Every rejection names the file, the column and the row. Nothing is guessed.

6. What already exists in the system, and what has to change

This section comes from a line-by-line inspection of the repository on 24 September 2026 [Analysis], made on the 20 August state of the branch (commit `a8c8926`) and then brought up to the branch head fetched the same day (commit `08485e2`, which adds 27 commits made between 19 August and 10 September). Every statement names the file that proves it.

6.1 The branch today, in plain terms

The August build put in place the bones of the engine and they work: files can be loaded through a mapping screen; fees can be classified as integral or not by one person and approved by another; an EIR can be solved and locked per contract; a monthly amortised-cost roll-forward can be run for a period; interest posted per account (Extract C) can be reconciled to EIR interest on screen; and the ECL module discounts at the solved EIR. The EIR and ECL test suites run and pass at the branch head: 170 tests, 24 skipped, 772 assertions.

Added in September (commits `ac030a6` to `af7c6e2`, 19 August to 2 September, plus the manuals of 10 to 13 September): a governed life for the version 1 schedule (a DRAFT generated from Extract A terms, compared with Extract B's remaining flows to within 1 percent, reviewed, APPROVED and then never

regenerated: `ScheduleWorkflowService`, `EirScheduleController`, `ScheduleShow.vue`, `contract_remaining_cashflow_schedule`, readiness blocker `SCHEDULE_NOT_APPROVED`); a date-sensitive solver on actual contractual dates under the contract's day count (`DateSensitiveEirService`), with admin-controlled reopening archived to `eir_calculation_history`; the trial-balance corpus (`gl_trial_balance_lines`, `gl_account_scope`, `TrialBalanceImportService`, `TrialBalanceMovementService`, `eir:import-trial-balances`, cumulative-YTD rule applied on read); a three-term variance bridge whose GL accrual base is derived from the data rather than assumed; a standing fee rulebook (25 seeded rules, priority-ordered sweep); time-phased, scenario-weighted ECL discounted at the EIR (`EclDiscountingService`, `TimePhasedEclService`, Projections screen); and the user, administrator, technical and installation manuals under `docs/manuals`.

Added on 25 September 2026 (phases P1 and P2 of the plan in 12.1, merged into the branch and pushed): the **Governance Centre** (`governance_settings` and `governance_setting_history`, `GovernanceService`, a screen, the twelve settings of section 8 seeded as approved defaults effective 1 January 2025, maker-checker on every change, the reconciliation band and the Stage 3 basis now read from it rather than from constants); the **reference-rate series** (`reference_rate_series`, an ISO-only importer that refuses an ambiguous or out-of-order file and names the row, `eir:import-reference-rates`, and a Reference Rates screen); the **verbatim E-Banker code columns** on `contract_eir` (Interest Policy, Floating Flag, calculation base, instalment basis, moratorium type and the rest of 6.3) loaded from Extract A, with `reprice_flag` derived from the policy and a note where the Floating Flag disagrees; `SpreadDerivationService` and `eir:derive-spreads`, which derive the spread added to the prime rate (margin) from the loan books and flag any account whose spread drifts; the `schemes` table (written in P4); **loan_books.repayments and commitments now populated** by the loan-book importer, which is what arrears will be read from; and the four EIR permissions. The suite after both merges: 233 passed, 24 skipped, 1,260 assertions.

What the branch still does not do is anything specific to a **floating reset**, a **moratorium of the "Principle Only" kind**, a **restructure**, or **partial disbursements**; and it accrues interest on a convention that is not E-Banker's. Those are the subjects of the phases that follow.

Tables in place (migrations `2026_07_27` to `2026_08_18`): `contract_eir` (59 attributes: identity, classification, pricing, dates, amounts, profile, results, solver audit, workflow, lineage), `contract_cashflow_schedule` (versioned, unique on contract + version + due date), `contract_fees` (signed lines, integral flag, maker-checker fields), `eir_amortisation` and `eir_amortisation_history`, `rate_reset_events` (present, never written or read), `eir_actual_transactions` (Extract B actual rows), `gl_interest_postings` (Extract C), `import_mappings`, `staging_thresholds`, `eir_accounting_rules`, `eir_fee_classification_events`.

Services in place (`app/Services/Eir/`): `CalculateEirService` (Newton-Raphson with bisection, ACADÉS golden test), `ScheduleGeneratorService` (annuity, one moratorium shape), `EirReadinessService`, `EirContractInputService`, `EirCalculationService` (orchestration and lock), `EirRevenueService` (monthly roll-forward, Stage 3 net basis), `EirGLReconciliationService`, `EirCoverageService`, the five import services (contract master, transactions, GL interest, schedules, fees), `FeeRuleMatcher`, `StagingClassifier`, `MappedFileReader`, and `EclDiscountRateService` for door 3.

Screens in place (`resources/js/Pages/Eir/`): Intake (upload, map, result), Calculations (solve, approve and lock), Data (contracts, cash flows, GL postings), Reconciliation (bridge tiles and per-facility table), Coverage (blockers by exposure), Fee Classification, Accounting Rules. Sidebar group "EIR and Revenue Recognition". The intake page is reachable only from a button on the Data page. Every EIR route sits behind the single broad permission `permission:settings`.

6.2 Capability map

Capability the engine needs	Today	Proof	Change
Solve an EIR by IRR, with an audit trail	Exists, date-sensitive since September	<code>CalculateEirService.php</code> , <code>DateSensitiveEirService.php</code>	Carry the same date basis into the revenue roll-forward
Fees netted off proceeds, maker-checker	Exists	<code>EirContractInputService.php</code> line 80	Respect <code>transaction_date</code> so a fee charged later in the life is not treated as at origination
Bulk import with column mapping and exception file	Exists	<code>MappedFileReader.php</code> , <code>ProcessEirImportJob.php</code>	Add the reference-rate, disbursement and restructure-schedule import types; ISO-only date rule
Monthly roll-forward with history, Stage 3 net basis	Exists, wrong convention	<code>EirRevenueService.php</code> lines 64-67 accrue $(1+EIR)^{(1/12)}-1$ per calendar month, no day count	Accrue at the EIR of the rate period in force, actual/365 (D9); look up the period's EIR from <code>eir_rate_periods</code>
Contractual interest as E-Banker posts it	Partial	<code>EirGlReconciliationService.php</code> derives the GL accrual base from the data since September (e553b48) but still models one rate per month with no day count	New <code>ContractualInterestService</code> : prior month-end balance x rate x days/365, first month from disbursement day inclusive, monthly capitalisation (7.7); the bridge reads it
Floating loans and PLR resets	Rate series built (P2); the reset itself still missing	<code>reference_rate_series</code> + <code>ReferenceRateImportService</code> + <code>ReferenceRate::inForce</code> exist; <code>rate_reset_events</code> still has no writer; revenue accrues at the locked EIR for every period	Reset detector; <code>eir_rate_periods</code> ; per-period solve (7.3), phase P5
Interest Policy, scheme settings, interest basis	Built (P2)	Verbatim columns on <code>contract_eir</code> , loaded by <code>ContractMasterImportService</code> ; <code>reprice_flag</code> derived from the policy; a note where the Floating Flag disagrees	Waiting on MAIIC to supply the codes (new ask 1); the <code>schemes</code> table is read in P4
Spread added to the prime rate (margin)	Built (P2)	<code>SpreadDerivationService</code> + <code>eir:derive-spreads</code> ; <code>spread_over_prime</code> , <code>spread_source</code> , <code>spread_drift_flag</code> on <code>contract_eir</code> ; the supplied <code>markup</code> is compared, not relied on	Run it on the nine monthly loan books once the reference series is loaded
Moratorium "Principle Only" versus "Both", grace period	One shape only	<code>ScheduleGeneratorService.php</code> lines 64-69 capitalise every moratorium; <code>ContractMasterImport.php</code> line 103 maps the principal grace into the same column	<code>moratorium_type</code> and <code>grace_period_months</code> columns; generator produces both shapes; the import conflict is removed
Interest on sanctioned amount versus balance	Missing	no column, no logic	<code>interest_calc_base</code> and <code>installment_based_on</code> columns; generator and contractual-interest service honour them (O6)

Capability the engine needs	Today	Proof	Change
Partial disbursements, undrawn commitments	Captured as text, not used	<code>disbursement_tranches</code> text; <code>CalculateEirService::validate</code> line 53 rejects a second drawdown; no code references <code>commitments</code>	<code>contract_disbursements</code> table; drawdowns enter the cash-flow vector as later outflows; undrawn = approved minus sum of drawdowns
Arrears from the loan book	Half built (P1)	<code>LoanBooksImport</code> now writes repayments and commitments	<code>LoanBookCashService</code> computes the monthly change and flags resets (D14, O5); B5.4.6 re-estimation in the roll-forward, phase P6
Original schedule governance	Exists since September	<code>ScheduleWorkflowService.php</code> , <code>ScheduleShow.vue</code>	Maker-checker on approval (O18)
Trial-balance control totals	Exists since September	<code>TrialBalanceMovementService.php</code>	Wire into the month-end run as test T6
Restructuring	Missing (version 1 is governed; version N+1 has no writer)	schedule version 2 has no writer (<code>ScheduleImportService.php</code> line 102 and <code>GenerateContractSchedules.php</code> line 108 hardcode 1); <code>modification_gain_loss</code> written as 0 (<code>EirRevenueService.php</code> line 81)	Version N+1 import; <code>contract_modifications</code> table; 10 percent test; lineage on sub-account (7.5)
Governance Centre	Built (P1)	<code>governance_settings</code> , <code>governance_setting_history</code> , <code>GovernanceService</code> , <code>Governance.vue</code> , twelve settings seeded; the reconciliation band and the Stage 3 basis read from it; a missing setting throws rather than defaulting silently	Point the remaining conventions at it as each phase touches them
Month-end run	Partial	<code>RunEirRevenueJob</code> runs a period for locked contracts; not scheduled, no screen, no period lock, no ordered pipeline; <code>RunEirRevenue.php</code> line 31 records a null user from the console	Ordered pipeline with a screen and a period lock (7.9); user recorded on every run
Exports	Missing	no download action on any EIR page (grep); the IFRS 9 hub has <code>Ifrs9ReportExport.php</code>	Excel and PDF for reconciliation, the revenue table and the audit pack, using the hub's export pattern
Single-contract view and edit	Missing	no route or form for <code>contract_eir</code>	Contract profile screen with rate periods, schedule versions, drawdowns and modifications
Permissions	Built (P1)	<code>eir.view</code> , <code>eir.run</code> , <code>eir.export</code> , <code>eir.govern</code> seeded; the read-only EIR screens and the Governance Centre now use them	Apply to the remaining routes as each phase touches them; note that the permissions seeder must run before anyone opens those screens

6.3 Data model changes

New tables (all with `created_by`, timestamps, and an `audit_logs` entry on every write):

Table	Grain	Columns	Keys
reference_rate_series	one row per rate change per index	index_code (PLR), effective_date, rate_decimal(8,5), source_row, as_delivered, interpretation, import_id	unique (index_code, effective_date); effective dates strictly increasing, enforced on import
schemes	one row per E-Banker scheme	scheme_code, product_code, interest_policy char(1), floating_flag char(1), interest_calc_base char(1), installment_based_on, emi_calc_type char(1), default_moratorium_type, effective_from	unique (scheme_code, effective_from)
eir_rate_periods	one row per contract per rate period	contract_id, sub_account_no, period_no, effective_from, effective_to null, reference_rate, spread_over_prime, contractual_rate, opening_amortised_cost, eir_period, eir_nominal_annual, eir_effective_annual, solver_method, solver_iterations, solver_residual, input_snapshot json, rate_reset_event_id, locked_at, locked_by	unique (contract_id, sub_account_no, effective_from)
contract_disbursements	one row per drawdown	contract_id, sub_account_no, tranche_no, disbursement_date, amount, reference, source_system, external_transaction_id	unique (source_system, external_transaction_id)
contract_modifications	one row per restructure	contract_id, sub_account_no, modification_date, old_schedule_version, new_schedule_version, carrying_amount_before, pv_new_flows_at_original_eir, change_pct, threshold_pct, treatment enum(MODIFICATION, DERECOGNITION), gain_loss, approved_by, approved_at	unique (contract_id, new_schedule_version)
eir_cash_receipts	one row per contract per month	contract_id, reporting_period, repayments_cumulative_open, repayments_cumulative_close, cash_received, source enum(LOAN_BOOK_DELTA, EXTRACT_B, SCHEDULE), reset_flag, reset_note	unique (contract_id, reporting_period)
governance_settings	one row per setting	key, value, options json, effective_from, set_by, approved_by, approved_at, reason	unique (key, effective_from)
governance_setting_history	append-only	copy of the row plus superseded_at, superseded_by	index (key)

Columns added to `contract_eir`: `scheme_code`, `interest_policy`, `floating_flag`, `interest_calc_base`, `installment_based_on`, `emi_calc_type`, `moratorium_type` enum(PRINCIPAL_ONLY, BOTH) stored beside the verbatim text, `grace_period_months`, `moratorium_from` date, `interest_start_date`, `first_instalment_date`, `spread_over_prime` decimal(8,5), `spread_source` enum(DERIVED, SUPPLIED), `spread_drift_flag`, `reprice_flag` (derived from `interest_policy`), `account_status_code`, `los_application_no`, `los_process_ref`, `predecessor_sub_account`. The existing `markup` keeps the value MAIIC supplies; `spread_over_prime` holds what the engine derived; the two are compared on screen.

Columns populated that exist but are empty: `loan_books.repayments` (the loan-book importer writes it from the report's Repayments column); `loan_books.commitments` (approved minus disbursed).

Tables used that exist but are idle: `rate_reset_events` (written by the reset detector, one row per contract per PLR change).

6.4 Repairs the inspection found

Items 2 and 5 were made in P1 on 25 September 2026 and are marked below. The rest stand.

1. Readiness is implemented twice (`EirReadinessService::assess` and `EirCoverageService::assess`; since September a test asserts the two agree) and reconciliation three times (the service, and raw SQL in `EirDataController` lines 68-115, MySQL-only). One implementation each; the controller calls the service.
2. **Done (P1).** `ContractMasterImport.php` mapped WEEKLY to 52 and FORTNIGHTLY to 26, which the readiness gate then rejected. They are now refused at import with a named reason. A numeric `payments_per_year` of 52 or 26 still reaches the gate.
3. The moratorium meaning conflicts between the import (principal grace) and the generator (full holiday with capitalisation). Resolved by `moratorium_type`.
4. `ScheduleImportService.php` line 55 refers to a restructure flow that does not exist. Built in 7.5.
5. **Done (P1).** `RunEirRevenue` now takes a required `--user=<id>`, checks the user exists, and refuses to run otherwise. `eir:import-reference-rates` follows the same rule.
6. The GL bridge assumed contractual/12 on the original drawn amount; September derives the base from the data. 7.7 completes it with the day-count rule and the disbursement-day start.
7. `docs/Development_of_EIR.md` says phases 4 to 7 are not started; they are. The log is brought up to date with this specification.
8. `storage/superseded-scratch-2026-08-04/` holds an abandoned schema. Deleted from the working tree (it is untracked) so no future `git add -A` picks it up.
9. `reporting_period` on `loan_books` is a free string read with `SUBSTR` and `REPLACE` tricks in three places. One helper normalises it.

7. How the numbers are worked out

This section explains each calculation the way it would be taught, using the figures from the teaching workbook [Teach] and from MAIIC's own data where a figure has been proven.

7.1 The contractual schedule

The engine first writes down what the borrower promised to pay, month by month, from the terms in File A. For a level-instalment (EMI) loan the instalment is the annuity payment on the drawn amount at the contractual rate over the remaining term. For an equal-principal loan the principal share is fixed and the interest share falls. The frequency can be monthly, quarterly or annual.

A moratorium changes the start of the schedule. "Principle Only" means the borrower pays interest but no principal for the moratorium months. "Both" means nothing is paid; the interest is added to the balance each month (D10) and the instalment is then sized on the enlarged balance. Both shapes are measured from the first disbursement date (E-Banker p.31). No third shape is built (D4).

Where E-Banker charges interest on the sanctioned amount rather than the balance (O6), the schedule follows that flag once MAIIC supplies it. On Lake Malawi Aquaculture (approved MWK 1,055 million, drawn 297 million, 34.75 percent) the two bases differ by MWK 22 million a month, so the flag is not optional.

7.2 The EIR at origination

The borrower receives the drawn amount less integral fees. In the teaching example a loan of 2,000,000 with 27,300 of arrangement and legal fees puts 1,972,700 in the borrower's hands. The EIR is the monthly rate that makes the present value of all the promised instalments equal to 1,972,700. The solver is Newton-Raphson with a bisection fallback, in payment-period units; the result is reported three ways and always labelled: the periodic rate (0.734967 percent a month in the example), the nominal annual rate (12 times that, 8.8196 percent) and the effective annual rate.

Which fees are integral is decided by the fee-classification rules already built (maker-checker, [Spec v2] phase 2.5). Only reviewed integral fees reach the solver.

7.3 Floating-rate resets (IFRS 9 B5.4.5)

When the Reserve Bank moves the prime rate, every loan with `Interest Policy = Link with PLR` reprices. The engine does three things at the effective date: it re-estimates the remaining cash flows at the new contractual rate (the new PLR plus the loan's unchanged spread over prime); it solves a fresh EIR on those flows against the carrying amount at that date; and it writes a new row in `eir_rate_periods`. There is no catch-up adjustment: the carrying amount does not jump. The previous EIR stays on record.

The spread over prime is derived (D13): the loan-book rate minus the PLR in force at that month-end. Over December 2025 to August 2026 the month-on-month change in rate on MAIIC Agricultural and Industrial loans equalled the PLR change exactly in every one of the eight months, across 17 to 37 accounts at a time, so the spread is constant per account and lands on clean values (4.90, 5.00, 6.00, 7.70 percentage points). FinES loans (0 of 74 moved while the PLR fell 4.5 points) and MAIIC Term Loans (0 of 12 moved) do not reprice. About 39 contracts do.

When a change falls inside a month, the timing rule is a governed setting (O2). The recommended rule is pro rata by days from the effective date, because that is how E-Banker accrues (7.7).

7.4 Arrears (IFRS 9 B5.4.6)

When an instalment is missed, the promised cash flows are no longer the expected ones. The engine keeps the EIR unchanged and recomputes the amortised cost as the present value of the **revised** expected flows at the **original** EIR; the difference goes to income in the month it is found. Interest keeps accruing at the EIR on the amortised cost, and the missed amounts stay in the expected flows until they are paid or written off.

The trap this engine is built to avoid: at the next reset date, if the cash-flow stream is no longer level (missed instalments now sit as a lump, or a moratorium has bent the stream), a closed-form annuity formula (Excel's RATE) gives the wrong answer. In [Teach] the loan misses instalments 10, 11 and 12 and cures at 15; at the third reset the true EIR by internal rate of return is 9.5502 percent nominal while RATE returns 9.0012 percent, an understatement of 55 basis points. The engine therefore always solves by IRR on the actual expected flows (D15).

Cash actually received is read from the loan book (D14): this month's cash equals this month's cumulative Repayments minus last month's, and that increase ties exactly to the fall in Outstanding Balance (Mach Milk: 41,750,000 to 45,750,000 to 48,750,000 to 51,250,000 to 53,650,000 across the months, each step matching the balance). Twenty-three accounts show the counter reset to zero in nine months; a reset is a discontinuity, never a negative payment, and is handled under O5.

Stage 3 accounts accrue on the net carrying amount (D11), with the stage read live from the ECL module.

7.5 Restructuring (IFRS 9 5.4.3 and B3.3.6)

A restructure gives the loan a new schedule. The engine loads it as `schedule_version` N+1; version 1 is never overwritten. It then runs the 10 percent test: the present value of the new cash flows at the original EIR, compared with the carrying amount before the change. If the difference is under 10 percent (O16), the loan is modified, not derecognised: the carrying amount is reset to that present value, the difference is a modification gain or loss, and the original EIR continues. In [Teach] the year-six restructure moves the carrying amount from 1,224,225.82 to 1,192,314.49, a 2.6 percent change, so it is a modification with a loss of 31,911.33; the revised EIR after the fee on modification is 8.0400 percent nominal.

E-Banker records a restructure as a new sub-account on the same account (vendor statement; the Reschedule Report, Menu ID 2839, lists them). The engine keys each contract on account plus sub-account and stores the predecessor so the lineage survives.

7.6 Partial disbursements and undrawn commitments

A facility can be drawn in tranches. Interest accrues on what has been drawn (or on the sanctioned amount, per O6), and the undrawn balance is a loan commitment reported separately under IFRS 9. At 31 August 2026 eight facilities carried MWK 3,472,435,397 undrawn, and the loan book's own Not Yet Disbursed column ties to approved minus disbursed to the kwacha. The engine needs one row per drawdown (date, amount, reference) in a small separate extract, because Extract A carries exactly one tranche per loan even where three were drawn.

7.7 The interest posting convention, proven against E-Banker

This is the finding that lets the engine reconcile to the general ledger. On a clean month, E-Banker's interest posting equals:

prior month-end outstanding balance x annual contractual rate x days in the month / 365

JVD Agro, April 2026: posted 4,692,858.90; reconstructed 4,692,859.03. May 2026: 4,889,144.85 against 4,889,144.82. Malawi Police SACCO, May 2026: 85,386,066.75 against 85,386,066.84. Dividing the annual rate by 12 instead is wrong by 8 to 9 percent every February.

In the month of disbursement, interest runs from the disbursement day **inclusive** to the month-end: Ebenezer Midian, drawn 8 August 2025, posted 6,418,801.92 against $314,900,900 \times 31 \text{ percent} \times 24/365 = 6,418,801.91$. Microloan Foundation, drawn 18 December 2025, ties to the cent on 14 days. A capitalising moratorium compounds monthly: the balance rises each month by exactly the interest posted.

Where the reconstruction did not match, the reason was always one of three and each names a data item: a loan paid out late in the month (needs the disbursement date); a catch-up posting (Promenade Medical Centre: nothing posted in November or December 2025, four months posted at once in February 2026); a tranche drawn mid-month (Milele: the month-end balance cannot match without the per-drawdown extract). Across the ten offer letters, 32 of 47 account-months are within 2 percent and every clean month is exact [Recon].

7.8 Reconciliation to the ledger (door 2)

Every month the engine produces, per loan: contractual interest (7.7), EIR interest (the amortised cost times the EIR for the period), and the difference. The contractual figure is reconciled to Extract C (or the GL-wise Interest Posted Report) per account, and the sum by GL code and month is reconciled to the trial balance. The difference between EIR and contractual interest is the proposed true-up journal (O10). Fees and interest cannot be attributed to a loan from the GL spools alone ([Spec v2] limitation 9 and 10), which is why Extract C at full coverage and a fee extract by account remain requests.

7.9 The month-end run

1. Load the Loan Book Report (ECL module) and any new File A/B/C/D rows.
2. Apply validation (5.7); stop on any rejection.
3. Detect reference-rate changes since the last run; open new rate periods for the affected loans (7.3).
4. For each loan: update expected cash flows (arrears 7.4, restructures 7.5, drawdowns 7.6); roll the amortised cost forward one month; compute EIR interest and contractual interest.
5. Reconcile (7.8); produce the exceptions list with a named reason per exception.
6. Present the run for review; on approval, lock the period. A locked period is never restated by a later settings change (D17).

8. The Governance Centre

Every convention the engine uses is a setting, not a line of code. The screen shows each setting with its options, the option in force, the effective date, who approved it and the full history. The defaults are Dupleix's recommendations; MAIIC can change any of them with an approver's sign-off, and the change applies from its effective date forward only.

Key	Setting, plainly	Options	Default (recommended)	Status
plr_mid_period	When a PLR change inside a month takes effect	pro rata from the effective date / next month-end / next instalment	pro rata from the effective date	O2, not yet agreed
reset_trigger	When the EIR is re-solved for a floating loan	at the PLR effective date / at the next instalment date / at month-end	at the PLR effective date	agreed (D8)
moratorium_capitalisation	How a "Both" moratorium compounds	monthly at posting / at the instalment frequency	monthly at posting	agreed (D10), proven
rate_source_precedence	Which source decides whether a loan reprices	Interest Policy, then observed behaviour, then product family / observed first / product family only	Interest Policy first	O3
margin_basis	Where the spread added to the prime rate (margin) comes from	LIVE_AT_DRAWDOWN / AS_CAPTURED / SUPPLIED	LIVE_AT_DRAWDOWN	O1, depends on the refresh-button answer

Key	Setting, plainly	Options	Default (recommended)	Status
stage3_interest_basis	Interest on Stage 3 loans	net carrying amount / gross with allowance unwind	net (5.4.1(b))	agreed (D11)
day_count	Day count	ACT/365 / 30/360	ACT/365	agreed (D9), proven
cash_source	Where actual cash received comes from	loan book (change in Repayments) / Extract B / contractual schedule	loan book	agreed (D14)
manual_policy_handling	Loans with Interest Policy = Manual	read the rate monthly and treat each change as a reset / treat as fixed	read monthly	O4
modification_threshold	The derecognition test	10 percent / other	10 percent	O16
recon_tolerance	When a difference is an exception	100 bp on the EIR and MWK 1 per account-month / amount threshold	100 bp and MWK 1	O11
counter_reset_handling	A Repayments counter that falls	discontinuity, cash from Extract B / settlement / manual	discontinuity	O5

Each change writes an audit row: setting, old value, new value, who changed it, who approved it, when, why (free text), and the first period it applies to. A locked period keeps the settings it was locked under.

9. Screens and the monthly workflow

9.1 Screens to add

Screen	Who uses it	What it shows and does
Governance Centre	Finance manager (govern), approver	Every setting in section 8 with its options, value in force, effective date, approver and history; propose a change with a reason; approve; a locked period shows the settings it was locked under
Reference Rates	Finance	The PLR series as a table and a step chart; import a new file; the 20 repaired rows with "what the file said, how we read it"; rejection messages for any ordering break
Contract Profile	Finance, auditor	One loan: terms as E-Banker holds them (codes verbatim, plain meaning beside each), the derived spread and the supplied spread side by side, drawdowns, schedule versions, rate periods with their EIRs, modifications, the amortisation roll-forward month by month, the interest reconciliation month by month with the named cause of each difference
Rate Resets	Finance	PLR changes detected since the last run, the loans affected, the new contractual rate per loan, the re-solved EIR, and the status (solved, blocked with reason)
Restructures	Finance (maker), approver	Load a version N+1 schedule for a loan; see the 10 percent test result; approve the modification; lineage to the predecessor sub-account

Screen	Who uses it	What it shows and does
Drawdowns	Finance	Import per-drawdown rows; undrawn commitment per facility and in total
Month-end Run	Finance (run), approver (lock)	The pipeline of 7.9 as numbered steps with status, counts and an exceptions list per step; run; review; lock the period
Exports	Finance, auditor	Excel and PDF for the reconciliation, the revenue table (EIR interest, contractual interest, difference, by loan and by stage), the audit pack (settings in force, rate periods, exceptions with causes)

Existing screens change as follows: Intake gains the three new import types and appears in the sidebar; Reconciliation gains the cause column and a download; Calculations shows the rate period being solved, not just the contract.

9.2 In-app help

Every new screen ships with its help-centre article and glossary entries in the same commit (`HelpContentSeeder.php`), written in the same plain language as this document. The user manual generator (Ticket #010) is re-run so the downloadable manual matches.

9.3 The monthly workflow, as the finance team will see it

1. **Load** the month's Loan Book Report (as today) and any new reference-rate, drawdown or restructure files. The Intake page shows what loaded, what was held and why.
2. **Check** the Rate Resets page: if the PLR moved, the affected loans are listed with their new rates and re-solved EIRs.
3. **Run** the Month-end Run: the pipeline executes; exceptions are listed with causes.
4. **Review** the Reconciliation: every loan where contractual interest differs from E-Banker's posting shows the cause (late disbursement, catch-up, tranche, data gap).
5. **Lock** the period with an approver's sign-off. Export the audit pack and the proposed journal.

10. Reconciliation, testing and acceptance

The contract makes acceptance turn on EIR calculations that "reconcile within the agreed tolerance". This section says what that will mean.

Test	What passes	Evidence today
T1 Interest reconstruction	On a clean month, contractual interest equals E-Banker's posting to the cent for every account; every difference has a named cause	32 of 47 account-months within 2 percent, clean months exact [Recon]; first-month accrual to the cent on 2 of 2 testable [ExtB]
T2 Offer-letter reproduction	The generated schedule reproduces the printed instalment on the offer letter	5 of 6 where the two rates differ (Milele's annual structure still to model)
T3 Workbook integrity	The engine reproduces every figure in [Teach] (EIR at origination, at each reset, after arrears, after modification; total EIR income 1,049,024.58 against contractual 982,313.25; final carrying amount 0.00)	28 of 28 checks pass in the workbook; engine fixtures to be written from it
T4 Spread constancy	The derived spread over prime is constant per account	23 of 35 stable to within 0.15 pp in the first pass; the rest quarantined

Test	What passes	Evidence today
T5 Cash tie-out	Change in Repayments equals the fall in Outstanding Balance	Exact on every non-reset month tested
T6 GL control total	Contractual interest summed by GL code and month equals the trial balance	Requires Extract C at full coverage (61 accounts missing)
T7 Test suite	The branch's PHPUnit suite is green on the in-memory database before and after the build	Green on 24 Sep 2026 at commit 08485e2 : 170 passed, 24 skipped, 772 assertions across the EIR and ECL suites
T8 Date integrity	No imported date is ambiguous	Importer rule (D19); [DateScan] on the loan books

11. Data requests: where each one stands

The full request-by-request record, with the wording of each email and the reason behind it, is in [Workbook], sheet "Requests". In summary, of the fifteen items of 11 September: three are closed (2 loan books, 3 take-on schedules, 14 manuals); three are partly received (7 rate history, 8 extra fields, 12 restructured loans); nine are outstanding, of which fees per facility (4), LOS schedules after go-live (5) and the EMI chart with a drawdown row (6) are critical. Two outstanding items are now less critical because the engine derives the spread (D13) and takes cash from the loan book (D14).

Six new asks were added on 24 September: Interest Policy and the other five scheme settings per account; the three documented reports never requested (Audit Trail 692, Reschedule 2839, IFRS9 Detail 3863); ISO dates on every extract and re-supply of Extracts A and B; the IFRS9 Detail Report's full column list and a sample; confirmation of the go-live date and a re-run of the Loan Book Report for November 2024 to November 2025; and the change request to extend the Loan Book Report. [Deck] slides 10 to 16 carry them.

The register behind the 11 September email, field by field with the source screen, the target column and the reporting period, is Appendix E of [Spec v2]. The three sample packs it requests (ten LOS-originated loans, ten migrated loans, and every restructured loan, each with its documents, schedules and screenshots) remain outstanding and are the fastest route to T1 and T2 at full coverage.

12. Build plan and who does what

12.1 Phases, in order

Phase	What is built	Depends on	Acceptance
P1 Foundations (delivered 25 Sep 2026)	Governance Centre tables, service and screen; the four permissions; repairs 6.4 items 2 and 5; loan_books.repayments and commitments populated	nothing	Met: 197 tests green at the merge, settings resolve by effective date, a missing setting fails closed, a change never restates an earlier period

Phase	What is built	Depends on	Acceptance
P2 Reference rates and spread (delivered 25 Sep 2026)	<code>reference_rate_series</code> import (ISO-only, ordering check); <code>SpreadDerivationService</code> with the constancy check; <code>schemes</code> and the verbatim code columns; <code>reprice_flag</code> ; Reference Rates screen	[PLR]; Interest Policy per account (new ask 1) or observed behaviour as fallback	Met in test: the repaired file loads as 48 rows, 26 rate changes, 4 Mar 2020 to 3 Sep 2026, current rate 21.20 percent; a dd/mm file and an out-of-order file are both refused. Still to run against the live loan books for T4
P3 Contractual interest and reconciliation	<code>ContractualInterestService</code> (7.7); reconciliation rewritten on it with named causes; exports	P2	T1: clean months to the cent on the ten sample loans
P4 Schedules	Generator: both moratorium shapes, grace period, EMI calc types, sanction versus balance basis, date-based periods; drawdowns table and import	P2; the six scheme settings per account	T2: offer-letter instalments reproduced (6 of 6 once Milele's annual structure is modelled); T3 workbook fixtures
P5 Floating resets	The Phase 5.1 order in [Spec v2] s.7.5, extended: readiness blocker <code>FLOATING_TERMS_MISSING</code> ; reset detector from <code>reference_rate_series</code> writing <code>rate_reset_events</code> ; reset intake with maker-checker; a reset inside a locked period refused unless the period is superseded with a reason; <code>eir_rate_periods</code> holding the spread-lock EIR and the re-solved EIR; revenue roll-forward reads the period's EIR and accrues actual/365; ECL discounts floating loans at the current EIR	P2, P3, P4; O17 confirmed by Deloitte before the first reset is booked	T3 reset figures from [Teach]; spread-lock equals re-solve within tolerance; a reset changes no prior period; every PLR-linked loan has one rate period per PLR change since its start date
P6 Arrears	<code>eir_cash_receipts</code> from the loan book; reset handling; B5.4.6 re-estimation in the roll-forward; IRR on actual expected flows	P4, P5	T5 cash tie-out; T3 arrears figures (the 55 basis-point case)
P7 Restructuring	Version N+1 import; <code>contract_modifications</code> ; 10 percent test; lineage	P6	T3 modification figures (2.6 percent change, loss of 31,911.33)
P8 Month-end run and screens	Pipeline, period lock, Contract Profile, Rate Resets, Restructures, Drawdowns, Month-end Run, help articles, manual regenerated	P1 to P7	A full month runs end to end on the Dec 2025 to Aug 2026 loan books with every exception explained
P9 Acceptance and UAT	T1 to T8 on MAIIC's data; UAT with the finance team	MAIIC data (section 11)	Contract milestone 3
P10 Deployment and training	Install in MAIIC's environment; training; manuals; source code handover	P9	Contract milestone 4, Go-Live, 90-day warranty

12.2 What each phase needs from MAIIC

Phase	Needs	Status
P2	The repaired PLR file confirmed (20 rows); Interest Policy and Floating Flag per account or the scheme code	File ready; confirmation and the codes outstanding
P3	Extract C for all accounts to the latest month; disbursement dates	61 accounts missing; dates in Extract B need ISO re-supply
P4	The six scheme settings per account; per-drawdown extract; File D where it exists	Outstanding
P6	Loan Book Reports monthly (have Dec 2025 to Aug 2026); explanation of counter resets	Nine months held; resets unexplained
P7	Reschedule Report; restructured loan pack	Four in the take-on workbook; the report never run
P9	Fees per facility; the fourteen-month gap decision (O8)	Outstanding; the largest remaining gaps

12.3 Who does what

Party	Responsibility
Dupleix, Kundai Muriwo	Builds P1 to P8 on branch <code>eir_revenue_recognition</code> , tests green at every commit, help articles in the same commit, commits only the files each change touches
Dupleix, Edward Mazibuko	Signs off each phase against the acceptance column; owns the conventions and this specification; leads the MAIIC meetings
MAIIC, Barry Makumba	Supplies the data in 12.2 as CSV with ISO dates; runs the standard reports; raises the Loan Book Report change request with Virtual Galaxy
MAIIC, Dr Thomson Kumwenda	Decides the open choices in section 4; approves the Governance Centre defaults; authorises the vendor change request; sponsors UAT
MAIIC, Finance team	Operates the monthly workflow from P8; UAT within five business days of notice (contract)
Virtual Galaxy	Prices and delivers the Loan Book Report extension (O9); answers how the rate file and the extracts were produced

12.4 The immediate build order

P1 and P2 were built on 25 September 2026 and are merged into `eir_revenue_recognition`, because neither waited on MAIIC. The next session starts with **P3**, the contractual-interest service and the reconciliation rewritten on it, because the convention is proven (7.7) and Extract C for 120 accounts is in hand; then **P4**, the schedule shapes, against the teaching workbook's fixtures so that it is ready when the scheme settings arrive. Two housekeeping items go with P3: run `eir:import-reference-rates` and `eir:derive-spreads` against the live loan books to complete test T4, and run the permissions and governance seeders on any environment before the EIR screens are opened. Nothing waits idle on a data request.

13. Risks and limitations

#	Risk or limitation	Effect	Mitigation
R1	The Excel date problem recurs on a future extract	Schedules shift by months; cash lands in the wrong period	ISO-only importer (D19); ask MAIIC never to re-save through Excel
R2	Interest Policy is not supplied	Repricing decided from observed behaviour and product family	O3 precedence; the Audit Trail Report as a fallback source
R3	The fourteen-month gap is not filled	EIR history for legacy loans starts at December 2025 or rests on Finance's Excel books	O8; disclose as a limitation in the audit pack
R4	The LOS schedule differs from the core cash flow (start date, rate)	Wrong expected flows if the schedule is trusted	O7; generate from core terms; keep the schedule as reference
R5	Tranche history and restructure lineage are lost on export	Interest on partly drawn loans cannot reconcile; modification accounting lacks its "before"	Per-drawdown extract; key on account plus sub-account
R6	Fees cannot be attributed per loan from the GL	Integral fees missing from the EIR for most loans	O14; interim mapping table disclosed
R7	Interest Policy = Manual accounts	No rule to follow	O4; read monthly from the loan book
R8	Repayments counter resets	Cash for that month unknown	O5; Extract B for those months
R9	Test suite not green	Regressions go unnoticed	D21 before any new build
R10	Governance settings changed without approval	Locked periods restated	Approver role; effective dating; audit log (D17)
R11	Real customer names and balances in the extracts	Confidentiality	Same care as all client data; never reused in training material

14. Manual pages and files cited

Item	Reference
Interest Policy (nine options)	E-Banker p.19 and p.27
Floating Flag (four options; Fixed is "Applicable for Time Deposit Account")	E-Banker pp.19-20 and p.28
Installment Based On	E-Banker p.22
Margin Period Required Flag (a grace concept, not an interest margin)	E-Banker pp.23, 31
Loan Interest Cal. Base On (N, S, B); EMI Calc Type (E, P, F)	E-Banker p.30
Moratorium Period, from the date of first disbursement	E-Banker p.31
Sanction types (Primary Appraisals Sanction, Enhancement of Limit, and others)	E-Banker p.12
Transaction Posting and the debit/credit indicator	E-Banker pp.58-60
Section 5 "MAIIC Report": Audit Trail 692, Reschedule 2839, IFRS9 Detail 3863, Loan Book 3868	E-Banker pp.75-77

Item	Reference
GL-wise Interest Posted Report figure (a production run for 1 to 31 October 2024)	E-Banker p.76
Moratorium Type: "Principle Only" and "Both (Interest + Principle)"	LOS p.76
Grace Period (In Month)	LOS p.77
Margin % on the collateral screen (collateral haircut)	LOS p.63
Funding Proposal: Interest Rate %, Margin %, Effective Rate %, Rate Type	LOS p.74
Repaired reference-rate series	[PLR]
Reconstruction and Extract B check	[Recon], [ExtB]
Loan-book date scan	[DateScan]
Teaching workbook	[Teach]

15. Glossary

Term	Plain meaning
Amortised cost	What a loan is carried at: the amount advanced, plus interest recognised, minus cash received, minus any impairment.
Contractual rate	The all-in rate on the repayment schedule: the reference rate plus the spread added to the prime rate (margin).
EIR, effective interest rate	The rate that exactly discounts the expected cash flows to the amount actually advanced net of integral fees. Slightly above the contractual rate whenever fees were deducted.
E-Banker	MAIIC's core banking system, by Virtual Galaxy. Holds the loan accounts and posts interest.
Extract A, B, C	The three hand-written queries MAIIC ran from E-Banker in August 2025: A is the loan account register at two dates; B is every posting to every loan in 2025; C is interest posted per loan per month.
Floating Flag	An E-Banker field (N, F, I, B) that says whether a rate may vary. Not the field that links a loan to the PLR.
Governance Centre	The screen where every calculation convention is set, approved, dated and logged.
Integral fee	A fee that is part of the cost of the loan (arrangement, legal) and so is netted off the amount advanced when solving the EIR.
Interest Policy	The E-Banker field (N, M, L, P, S, U, R, W, F) that says where a loan's rate comes from. P = Link with PLR; F = Fixed; M = Manual.
IRR, internal rate of return	The discount rate that makes the present value of a set of cash flows equal to a given amount. The EIR is an IRR.
ISO date	A date written year-month-day, 2026-09-24 . It cannot be misread under any country setting.
Loan Book Report	E-Banker Menu ID 3868, the monthly 29-column report of every loan account. A custom report Virtual Galaxy built for MAIIC.
LOS	The Loan Origination System, where applications and offer letters are prepared before the account is created in E-Banker.
Menu ID	The number that identifies a screen or report in E-Banker.

Term	Plain meaning
Modification	A change to a loan's terms that is not large enough to count as a new loan; the original EIR continues and a gain or loss is booked.
Moratorium	A period at the start of a loan when principal (Principle Only) or both principal and interest (Both) are not paid.
PLR	The Reserve Bank of Malawi's prime lending rate, the reference rate for MAIIC's floating loans. 26 changes since March 2020.
Rate period	The stretch between two reference-rate changes during which one EIR applies to a floating loan.
Reference rate	The PLR. The rate typed in the body of an offer letter.
Reset	The date a floating loan's rate changes because the reference rate changed.
Sanction, sanctioned amount	The approved facility amount, which may be more than what has been paid out.
Scheme	E-Banker's level between product and account. The six settings that decide rate behaviour and interest basis live here.
Spread added to the prime rate (margin)	The fixed percentage MAIIC adds to the PLR to arrive at the contractual rate. Constant per loan. Never called just "margin" in this document.
Stage 1, 2, 3	The IFRS 9 credit-risk stages: performing; significantly deteriorated (30 days past due backstop); credit-impaired (90 days backstop).
Sub-account	E-Banker's numbering of a restructured loan under the same account number.
Tranche	One drawdown of a facility that is paid out in parts.
Undrawn commitment	The approved amount not yet paid out. Reported separately under IFRS 9.